MITCHELL GONDWE

BEng Computer and Electronic Engineering Student

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PROFESSIONAL SUMMARY

Final-year Computer and Electronic Engineering (BEng) student at North-West University with hands-on experience bridging embedded firmware, hardware instrumentation, and wireless systems. Practical background spanning C/C++ development on STM32 microcontrollers, peripheral interfacing (UART, SPI, I2C, ADC), and bench testing with oscilloscopes, logic analysers, and LTspice, alongside industry exposure to cellular IoT, GNSS, and LoRaWAN. Driven by a disciplined approach to physical debugging, hardware bring-up, and taking systems from schematic to verified, reliable operation.

SKILLS

Technical Proficiency: C, C++, Java, circuit design

Team Collaboration: Works effectively in cross-functional project teams

Adaptability: Quick to learn and apply new technologies and working environments

Community Engagement: Active volunteer in engineering outreach programmes

WORK EXPERIENCE

Vacation Work Trainee

Otto Wireless Solutions (Pty) Ltd, Randburg | June 2026 – July 2026

- Completed a five-week vacation work training programme covering IoT, LoRaWAN, RF and wireless technologies, cloud connectivity, and industrial networking.
- Gained hands-on experience in product development, system integration, and the creation of technical documentation.
- Deepened understanding of practical applications for wireless communication and IoT technologies.

Student Assistant

School of Computer Science and Information Systems, NWU | First Semester, 2024 – 2026

- Graded tests and assignments, invigilated examinations, and captured and uploaded marks for a C programming module, ensuring accurate assessment of student performance.
- Supported students with programming-related queries, reinforcing their understanding of key programming concepts through targeted assistance.

Temporary Assistant

Mechanical Engineering Department, NWU | First Semester, 2024 – 2026

- Marked Engineering Graphics assignments using Siemens NX, ensuring accuracy and adherence to technical drawing standards.
- Evaluated students' CAD models and technical drawings, providing constructive feedback to enhance their understanding of engineering design principles.
- Supported academic and administrative functions within the Mechanical Engineering Department, contributing to overall departmental efficiency.

EDUCATION

BEng Computer and Electronic Engineering

North-West University |02/2022 – 12/2026

Latest results: Databases and Web Programming 77% | Telecommunication Systems 70% | Control Theory II 69%|Embedded operating systems 61% |Signal theory 64% |Engineering management 60%

National Senior Certificate (Bachelor's Pass)

Victory House Private School |01/2016 – 12/2020

Matric results: Accounting 89% | Life Orientation 85% | Life Sciences 80% | English 74% | Mathematics 70% | Afrikaans 70% | Physics 70%

FINAL YEAR PROJECT

Android-Based Condition Monitoring for Induction Motors
BEng Mini-Dissertation, North-West University | 2026

- Designing a standalone Android application performing on-device, real-time vibration-based condition monitoring of induction motors, removing the need for internet connectivity or backend infrastructure, addressing a key limitation in remote agricultural and industrial environments.
- Implement a JPCCED-HI statistical health-index model, extracting 18 time- and frequency-domain features and applying Principal Component Analysis to classify motor condition and diagnose one of seven failure modes.
- Build a hybrid Kotlin + C++ (NDK) architecture, using C++ for computationally intensive signal processing (FFT, Hann windowing, feature extraction) to meet strict real-time processing requirements, achieving 290–665ms execution on mid-range hardware.

LANGUAGES

English: Fluent

Afrikaans: Intermediate

EXTRACURRICULAR ACTIVITIES

Facilitate workshops at Engenius Kids, a programme introducing engineering concepts to Grade 9 learners in township schools through engaging, hands-on projects, with the aim of encouraging continued study of Pure Mathematics and Physical Science.

REFERENCES

Dr Rodney Sebopelo
Senior Lecturer, North-West University

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